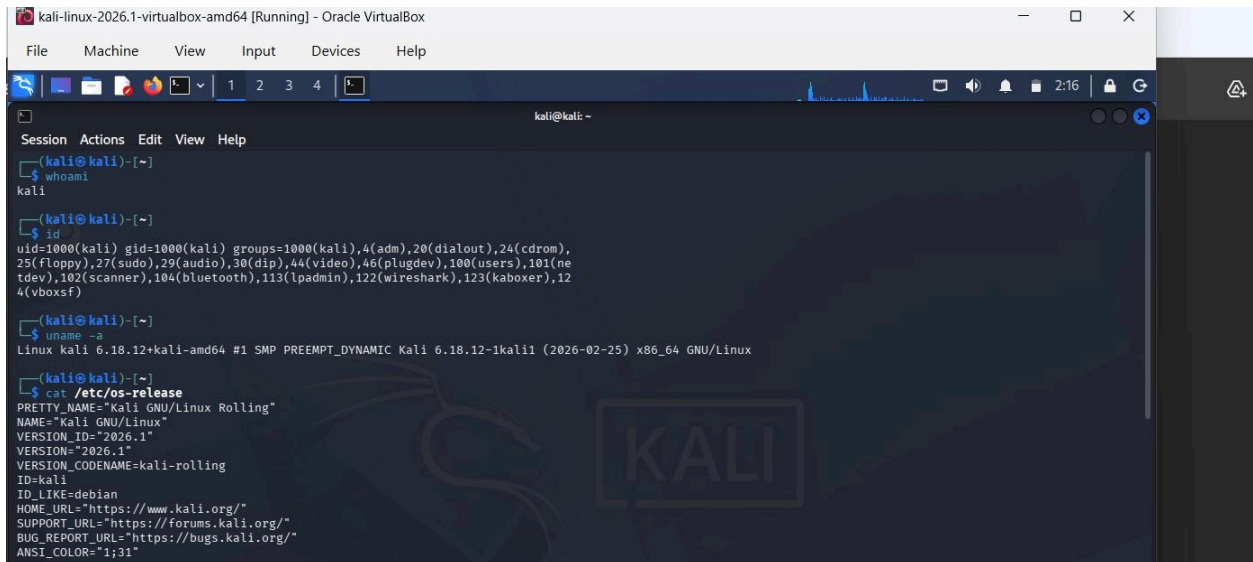


System Reconnaissance

Objective: Identify your identity, system details, and existing users

Deliverable: Screenshot showing the output of all the above commands. Flag:

GS{LINUX_RECON_MASTER}

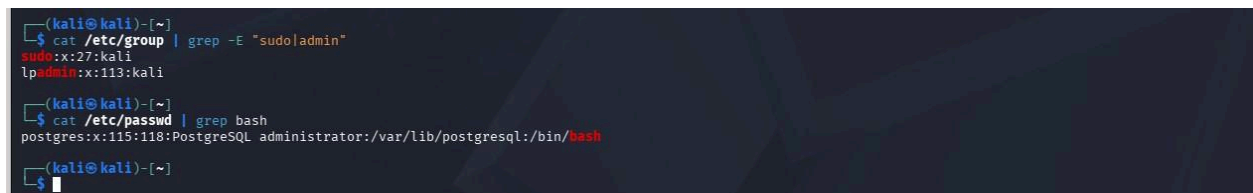


```
kali@kali:~$ whoami
kali

kali@kali:~$ id
uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom),
25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(ne
tdev),102(scanner),104(bluetooth),113(lpadmin),122(wireshark),123(kaboxer),12
4(vboxsf)

kali@kali:~$ uname -a
Linux kali 6.18.12-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64 GNU/Linux

kali@kali:~$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.1"
VERSION="2026.1"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"
```



```
kali@kali:~$ cat /etc/group | grep -E "sudo|admin"
sudo:x:27:kali
lpadmin:x:113:kali

kali@kali:~$ cat /etc/passwd | grep bash
postgres:x:115:118:PostgreSQL administrator:/var/lib/postgresql:/bin/bash

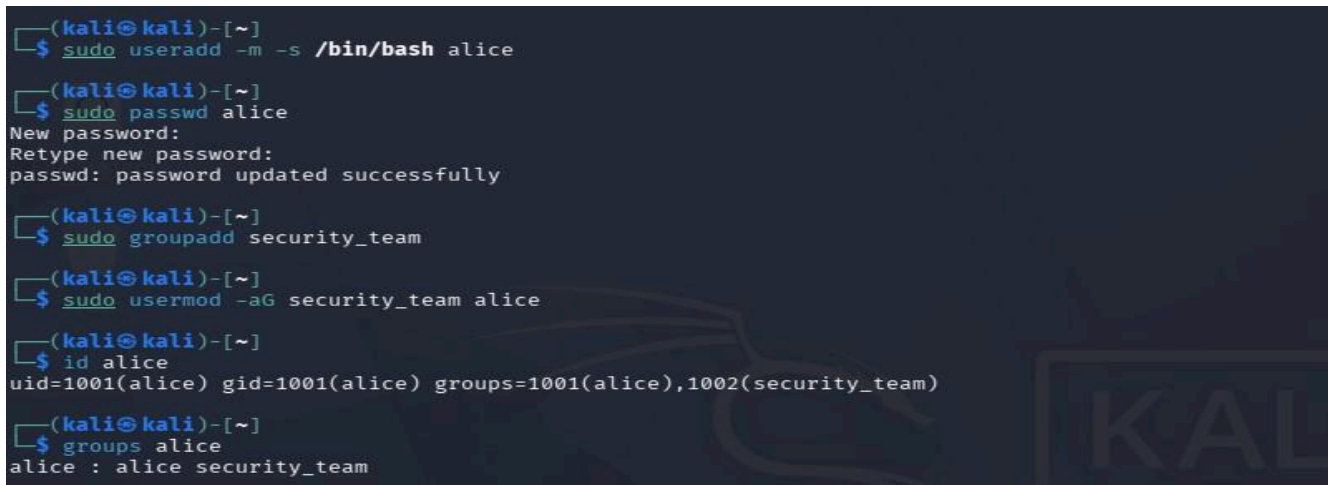
kali@kali:~$
```

Create and Manage Users

Objective: Create a new user account for incident response purposes.

Deliverable: Screenshot showing the user creation and verification steps.

Flag: GS{USER_GROUP_ALICE_SECURE}



```
kali@kali:~$ sudo useradd -m -s /bin/bash alice

kali@kali:~$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully

kali@kali:~$ sudo groupadd security_team

kali@kali:~$ sudo usermod -aG security_team alice

kali@kali:~$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),1002(security_team)

kali@kali:~$ groups alice
alice : alice security_team
```

```
(kali@kali)-[~]
$ sudo -l -U alice
User alice is not allowed to run sudo on kali.
```

Process Intelligence

Objective: Investigate running processes on your Kali VM.

Deliverable: Document the answers with screenshots of the commands used. Flag:

GS{PROCESS_SSH_ROOT_ANALYSIS}

```
(kali@kali)-[~]
$ ps aux --sort=-%cpu | head -10
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	626	2.9	7.0	436256	143016	tty7	Ssl+	03:03	2:19	/usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7
-novtswitch										
kali	1140	0.9	2.5	301040	52460	?	Sl	03:03	0:44	/usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /usr/lib/x86_64-linux-gnu/xfce4/panel/plugins/libcpugraph.so 13 25165836 cpugraph CPU Graph Graphical representation of the CPU load
kali	964	0.6	0.1	280916	3344	?	Sl	03:03	0:31	/usr/bin/VBoxClient --draganddrop
kali	2389	0.6	3.4	663152	69912	?	Sl	03:04	0:28	/usr/bin/qterminal
kali	1142	0.5	1.5	277064	30412	?	Sl	03:03	0:27	/usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /usr/lib/x86_64-linux-gnu/xfce4/panel/plugins/libgenmon.so 15 25165838 genmon Generic Monitor Show output of a command.
kali	1033	0.5	7.0	1178816	141716	?	Sl	03:03	0:26	xfwm4
kali	2432	0.2	0.3	13904	7544	pts/0	Ss	03:04	0:12	/usr/bin/zsh
root	1255	0.2	0.5	318792	10752	?	Ssl	03:03	0:10	/usr/libexec/upowerd
message+	484	0.2	0.3	9952	6732	?	Ss	03:03	0:09	/usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-ac
tivation										--syslog-only

What process is using the most CPU?

=> /usr/lib/xorg/Xorg is the process using the most CPU

```
(kali@kali)-[~]
$ sudo ss -tulnp
```

```
[sudo] password for kali:
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
tcp	LISTEN	0	128	0.0.0.0:22	0.0.0.0:*	users:(("sshd",pid=65144,fd=6))
tcp	LISTEN	0	128	:::22	:::*	users:(("sshd",pid=65144,fd=7))

What services are listening on network ports?

=> SSH-Port no 22 listening on network ports.

Log Analysis – Find the Intruder

Objective: Simulate failed login attempts and analyze the logs

Deliverable: Screenshot showing at least 5 failed login attempts in auth.log. Flag:

GS{FAILED_LOGIN_192_168_1_105}

```

2026-05-06T06:25:03.496397-04:00 kali sshd-session[82756]: Failed password for invalid user fakeuser from ::1 port 50158 ssh2
2026-05-06T06:25:10.074298-04:00 kali sshd-session[82756]: pam_unix(sshd:auth): check pass; user unknown
2026-05-06T06:25:12.363504-04:00 kali sshd-session[82756]: Failed password for invalid user fakeuser from ::1 port 50158 ssh2
2026-05-06T06:25:17.337298-04:00 kali sshd-session[82756]: pam_unix(sshd:auth): check pass; user unknown
2026-05-06T06:25:19.043395-04:00 kali sshd-session[82756]: Failed password for invalid user fakeuser from ::1 port 50158 ssh2
2026-05-06T06:25:19.520412-04:00 kali sshd-session[82756]: Connection closed by invalid user fakeuser ::1 port 50158 [preauth]
2026-05-06T06:25:19.527440-04:00 kali sshd-session[82756]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1
2026-05-06T06:25:27.700388-04:00 kali sshd-session[83070]: Invalid user fakeuser from ::1 port 57048
2026-05-06T06:25:29.996585-04:00 kali sshd-session[83070]: pam_unix(sshd:auth): check pass; user unknown
2026-05-06T06:25:29.996756-04:00 kali sshd-session[83070]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1
2026-05-06T06:25:31.848675-04:00 kali sshd-session[83070]: Failed password for invalid user fakeuser from ::1 port 57048 ssh2
2026-05-06T06:25:34.915858-04:00 kali sshd-session[83070]: pam_unix(sshd:auth): check pass; user unknown
2026-05-06T06:25:36.908003-04:00 kali sshd-session[83070]: Failed password for invalid user fakeuser from ::1 port 57048 ssh2
2026-05-06T06:25:39.652190-04:00 kali sshd-session[83070]: pam_unix(sshd:auth): check pass; user unknown
2026-05-06T06:25:41.479977-04:00 kali sshd-session[83070]: Failed password for invalid user fakeuser from ::1 port 57048 ssh2
2026-05-06T06:25:41.834490-04:00 kali sshd-session[83070]: Connection closed by invalid user fakeuser ::1 port 57048 [preauth]
2026-05-06T06:25:41.837429-04:00 kali sshd-session[83070]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1
2026-05-06T06:26:14.675882-04:00 kali sudo:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tail -20 /var/log/auth.log

(kali@kali)~$
$ sudo grep "Failed password" /var/log/auth.log | tail -5
2026-05-06T06:25:19.043395-04:00 kali sshd-session[82756]: Failed password for invalid user fakeuser from ::1 port 50158 ssh2
2026-05-06T06:25:31.848675-04:00 kali sshd-session[83070]: Failed password for invalid user fakeuser from ::1 port 57048 ssh2
2026-05-06T06:25:36.908003-04:00 kali sshd-session[83070]: Failed password for invalid user fakeuser from ::1 port 57048 ssh2
2026-05-06T06:25:41.479977-04:00 kali sshd-session[83070]: Failed password for invalid user fakeuser from ::1 port 57048 ssh2
2026-05-06T06:27:06.138612-04:00 kali sudo:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/grep "Failed password" /var/log
.log
(kali@kali)~$

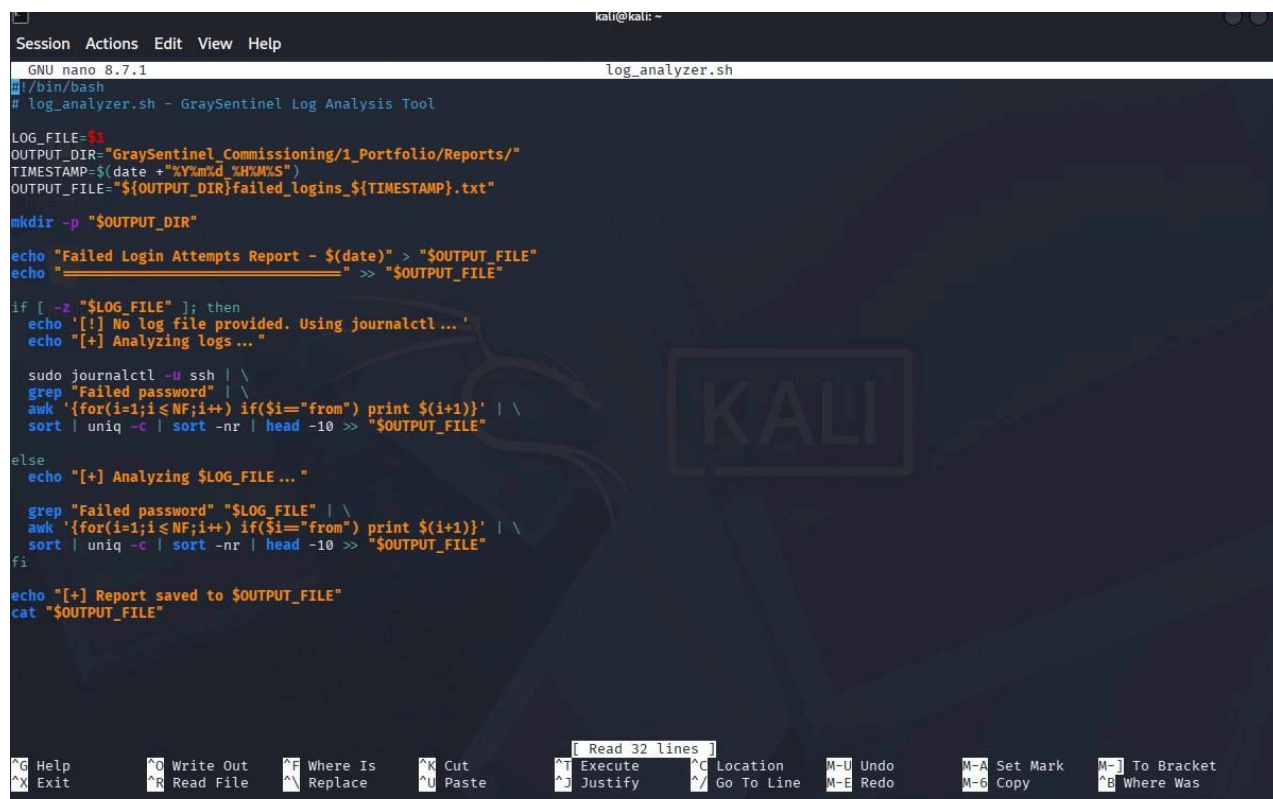
```

Create Log Analysis Automation

Objective: Write a Bash script that automates the extraction of top 10 IPs with failed login attempts.

Deliverable: The script file and a screenshot showing its execution with output. Flag:

GS{AUTOMATION_LOG_ANALYZER}



```

kali@kali: ~
Session Actions Edit View Help
GNU nano 8.7.1 log_analyzer.sh
#!/bin/bash
# log_analyzer.sh - GraySentinel Log Analysis Tool

LOG_FILE=$1
OUTPUT_DIR="GraySentinel Commissioning/1_Portfolio/Reports/"
TIMESTAMP=$(date +"%Y%m%d_%H%M%S")
OUTPUT_FILE="${OUTPUT_DIR}failed_logins_${TIMESTAMP}.txt"

mkdir -p "$OUTPUT_DIR"

echo "Failed Login Attempts Report - $(date)" > "$OUTPUT_FILE"
echo "===== " >> "$OUTPUT_FILE"

if [ -z "$LOG_FILE" ]; then
    echo "[!] No log file provided. Using journalctl..."
    echo "[+] Analyzing logs..."

    sudo journalctl -u ssh | \
    grep "Failed password" | \
    awk '{for(i=1;i<NF;i++) if($i=="from") print $(i+1)}' | \
    sort | uniq -c | sort -nr | head -10 >> "$OUTPUT_FILE"
else
    echo "[+] Analyzing $LOG_FILE..."

    grep "Failed password" "$LOG_FILE" | \
    awk '{for(i=1;i<NF;i++) if($i=="from") print $(i+1)}' | \
    sort | uniq -c | sort -nr | head -10 >> "$OUTPUT_FILE"
fi

echo "[+] Report saved to $OUTPUT_FILE"
cat "$OUTPUT_FILE"

```

```

(kali@kali)-[~]
$ sudo ./log_analyzer.sh /var/log/auth.log

+] Analyzing /var/log/auth.log ...
+] Report saved to GraySentinel_Commissioning/1_Portfolio/Reports/failed_logins_20260506_081821.txt
Failed Login Attempts Report - Wed May  6 08:18:21 AM EDT 2026
=====
18 ::1

```

File Integrity Monitoring

Objective: Set up file integrity monitoring for critical system files.

Deliverable: Screenshot showing the hash mismatch after modification. Flag:

GS{INTEGRITY_HASH_MISMATCH}

```

(kali@kali)-[~]
$ sudo sha256sum /etc/passwd /etc/shadow /etc/sudoers > baseline_hashes.txt

(kali@kali)-[~]
$ echo "# GraySentinel Audit" | sudo tee -a /etc/passwd
# GraySentinel Audit

(kali@kali)-[~]
$ sha256sum /etc/passwd | diff baseline_hashes.txt -
1,3c1
< 60939023dfaa5662319100673f5e550313bbf8167528a5a3ddff6dc430a70e80 /etc/passwd
< f26af12718d11c9b01e5cbfd1582bb399940fee5d896dde36466ede6ec21555 /etc/shadow
< 3e39081d74d699e8af589f87fbc681ee58523f901cdd7ef7ef3e2996b5bd0921 /etc/sudoers
> 87a3f54ea9903eda6e8b156742fed7be956768ba2f03e3a369692bba76069a7 /etc/passwd

```